Soham Desai

Third Year B.Tech · AI & Machine Learning

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PERSONAL PROFILE

Third-year B.Tech student in Artificial Intelligence and Machine Learning, with a minor in AI & Data Science from IIT Mandi. I build machine learning models, backend services, mobile apps, Unity games and drone hardware.

Previously a Software Development Intern at Sypha.AI, working on backend services, prompt optimisation and AI code analytics. I also build web platforms for freelance clients. Seeking internships in AI, Machine Learning and Software Development.

SKILLS

- **Languages:** Python, C++, C#, Java, JavaScript, TypeScript, SQL
- **ML & Data:** PyTorch, Hugging Face Transformers, scikit-learn, Pandas, PySpark, Spark MLlib, Power BI
- **Frameworks:** Flask, FastAPI, React, Spring Boot, React Native, Expo, PySide6, Unity
- **Databases:** PostgreSQL, MySQL, Supabase, Firebase
- **Cloud & Tools:** AWS (Lambda, API Gateway, DynamoDB, S3), Azure OpenAI, Git, GitHub Actions, Docker, Blender, Arduino IDE

ACADEMIC PROFILE

Atlas Skilltech University JUL'24 – JUN'28
B.Tech, Artificial Intelligence & Machine Learning

IIT Mandi COMPLETED
Minor in AI & Data Science: Data Science, Machine Learning, Deep Learning

Aditya Birla World Academy AUG'22 – FEB'24
AS / A Levels

CERTIFICATIONS

- Introduction to Cybersecurity – Cisco Networking Academy (Oct 2026)
- Minor in AI & Data Science – IIT Mandi (3 of 3 courses)
- Unity 2D & 3D Game Development, Game Physics in Unity, Blender 3D Modelling – Udemy

ACHIEVEMENTS

- Shortlisted – Smart India Hackathon 2024
- 1st (2019) and 3rd (2018) Place – Game Design, Design Championship India
- 5th Place Nationally – FIRST LEGO League Robotics
- Subject Topper – ICT (10th) and Computer Science (12th)
- State Champion, Volleyball · 3rd Place, Interstate Football

EXPERIENCE

Software Development Intern – Sypha.AI SEP'25 – FEB'26

- Worked on an AI-powered coding assistant. Integrated the Razorpay payment gateway, with backend workflows for subscription management and transaction verification.
- Built a prompt analysis module that evaluates and refines user prompts to make generated code more accurate.
- Built an AI code adoption analytics system measuring how much generated code users keep as-is versus modify.
- Developed Python and Flask REST APIs and structured data pipelines in an agile, Git-based team.

Freelance Developer

- **Imperio Selective Systems:** window configurator and quoting platform with bulk project uploads and phone-verified proforma quotes.
- **Kala Vyasanga** (in progress): curated marketplace connecting artisans' handmade products with buyers.

PROJECTS

PatraRakshak (team project; PyTorch, MuRIL, FastAPI): Finds prompt injections hidden in text and PDF/DOCX files, in English, Marathi and Hindi. Built the data, evals, API and UI; shipped an ensemble that cut false alarms from 14.5% to 4%.

Forge (React Native, Expo, TypeScript, Supabase): Fitness and health tracker with workout logging, weekly and cycle scheduling, analytics, and offline food and calorie tracking.

Protein Structure Prediction (PyTorch): Predicts 8-state protein secondary structure from amino-acid sequences using multi-kernel CNN layers feeding a GRU.

Lights Out (Python, PySide6, FastF1): Formula 1 companion for Windows: desktop widgets, a live dashboard and a pit-strategy simulator calibrated on 200+ races.

Resonance (team project): AI music recommendation platform. Built the micro-location tracking that feeds the recommendation engine.

ShopAI (team project; AWS, Azure OpenAI): Multi-vendor marketplace. Moved the backend to AWS Lambda, API Gateway, DynamoDB and S3; built the AI shopping chat.

MedVault (Spring Boot, React Native, MySQL): Medical records shared between patients, doctors and lab technicians through 5-minute QR access tokens.

Steam × Twitch Analytics (PySpark, Spark MLlib, Power BI): Joins Steam reviews with Twitch viewership, scores trending games and classifies popularity.

Other: Glitch Runner (Unity puzzle-platformer built on deliberate glitches), 4 more Unity games, 3 p5.play games, React Native + Firebase apps, Arduino builds.

POSITIONS OF RESPONSIBILITY

- **Hardware & Software Head:** International Space Drone Championship – BITS Goa, Feb 2025
- **Head of Esports:** SPARC Society – Atlas Skilltech University
- **Coding Mentor:** Python, Unity and programming basics
- **Beach Clean-up Team Leader:** 1 year leading, 3 years in total